

LabelProof Testing and Acceptance Plan

Document LP TEST 1.0 | Prepared 6 October 2026 | Product baseline 3.2 | Delivery package 3.3

Test objective and evidence policy

Verify that users can capture a product, inspect uncertainty, correct extraction, retain assessment history, export evidence and use private cloud workflows without misleading conclusions or cross-tenant access. This plan separates executed prototype checks from unexecuted production targets. A green unit test is not a legal approval, a general OCR accuracy result or proof that email reaches a real inbox.

The release record must identify source commit, app version, environment, test command, fixture provenance, result and unresolved limits. Tests involving cloud mutations use operator-provisioned disposable accounts; public demo tests use the read-only synthetic workspace. Never use customer credentials as test data or send email without an authorized test inbox.

Test environments and setup

Use npm ci from the committed lockfile and a supported Node release; development has used Node 24. Install or provide Chrome for the existing Playwright channel configuration. TEST_URL selects the frontend origin. Vite serves local frontend tests; Vercel serves real /api metadata checks. A Vite-only server must not be mistaken for local serverless API support.

The local full-test runner starts a fresh Vite process, waits for readiness, runs the selected suites and stops its own process. Synthetic image fixtures must live inside the repository or be generated by the test. Scratch downloads remain under ignored work/. The deployment test points to the live alias. Demo integration reads public configuration from that alias and intentionally public demo credentials.

Normal cloud regression requires an ignored .env.cloud-test and three disposable confirmed example.test accounts provisioned by an authorized operator. Set up isolated organizations and images, capture results and clean up those test resources after completion. No normal cloud result should be claimed for a run that lacked these accounts.

Executable suite inventory

- npm test: unit and metadata contract checks for uncertainty, review validation, evidence links, bounds, backup rejection, CSV safety and format-check limits.
- tests/startup.mjs: hold cloud configuration requests and prove the local dashboard and documents render before the cloud request completes. Cloud delays must not block saved local evidence.
- npm run test:buttons: interaction checks covering PDF export, zoom/filter controls, targeted rescan, evidence dialog, notifications, import rejection, cancellation, history recovery and mobile navigation dismissal.
- npm run test:upgrades: account/team/processing routes, comparison, local preliminary review, saved capture and recoverable queued batches.
- npm run test:storage: simulate a legacy database connection, blocked upgrade, completion after the old connection closes and preserved settings.
- npm run test:ocr: real Tesseract extraction, conflicting price rescan, historic report, image persistence and backup restoration using clean synthetic labels.

- npm run test:ocr-errors: invalid OCR input and pre-cancelled worker handling against local Vite source modules.
- npm run test:documents: documentation navigation, manifest, actual downloadable PDF/Markdown/prompt/ZIP assets and mobile controls.
- npm run test:deployment: live routes, evidence image loading, release version and real metadata APIs.
- npm run test:demo: real public-demo password auth, one visible synthetic organization, three products, write denial, credential/MFA protection, browser download, sign-out and mobile layout.
- tests/cloud.mjs: separately provisioned normal-account isolation, role enforcement, review binding, private image restoration, stale sync and cross-device consistency.

Requirement traceability and cases

TC 001 Capture and validation

Maps FR 001 to FR 005. Create a named product with a valid PNG, set its surface, reload the draft and confirm metadata and image persist. Reject a text file renamed as an image, wrong signature, oversized file, excess count and image above the pixel limit. Removing a draft image changes only the draft. Expected result: no invalid image reaches processing or a committed assessment.

TC 002 Safe observation states

Maps FR 006 to FR 010 and NFR 002. Use a product with no relevant photograph, a deliberately unreadable image and a readable image without detected text. Expect Not photographed, Unreadable and Needs review respectively. None becomes Potential issue. Verify net quantity does not use serving size. An observed declaration retains the correct source.

TC 003 Real OCR and conflicts

Maps FR 006 and FR 009. Process a clear synthetic English label with known fields. Rescan with a different MRP. Expect both values and images in Conflicting, with old version unchanged. Export and restore original image bytes. Record this as a fixture-specific functional test; do not report it as multilingual real-world accuracy.

TC 004 Correction prerequisites

Maps FR 012 and FR 013. Reject observed text without evidence, absence without applicability, coverage or rationale, non-applicability without a reason and boxes outside the original dimensions. Accept a valid reviewed correction and append a version. Editing the value without supplying a new valid region clears its old highlight.

TC 005 History and exports

Maps FR 014 to FR 016. Reopen every assessment after rescan, review and import. Earlier versions are read-only. Inspect PDF metadata and every evidence page; parse JSON and CSV values. Formula-leading text remains safe. Backup import rejects dangling image references, remote URLs, malformed schema and unreviewed absence. Repeated cloud pulls update matching records without duplicating products.

TC 006 UI controls and navigation

Maps FR 029 and NFR 004. Exercise every primary, secondary, modal, filter, file and navigation control by its accessible name or stable selector. Verify the resulting state, not only the absence of an exception. At 390-pixel width, dismiss navigation by button, backdrop, Escape and a destination. Check long names, empty lists, historical-version failure and reload of deep links.

TC 007 Demo restrictions

Maps FR 017 and FR 021. Sign in using public credentials; expect exactly the demo organization's permitted data. Attempt direct RPC writes, image upload, self-promotion, credential changes and MFA enrollment. Expect server denial even if client controls or user_metadata are forged. Original credentials continue to work. Browser sign-out affects only the current demo session.

TC 008 Normal cloud authorization

Maps FR 018 to FR 020 and NFR 001. Use two members and one outsider. Anonymous and outsider requests cannot read snapshots, private images or review events. An uploader cannot promote itself or approve; a reviewer can record allowed decisions; an admin can manage members. Review records bind exact snapshot version and assessment hash. A stale expected version fails without overwriting.

TC 009 Recoverable processing

Maps FR 004, FR 006 and FR 011. Cancel browser OCR and verify the draft remains usable. Close or reload during batch work, then retry the unfinished case without duplicate results. Target remote tests additionally kill a worker after lease acquisition, deliver a message twice, exhaust retries, cancel between stages and recover from provider unavailability. These remote cases remain pending.

TC 010 Public account email

Maps FR 022. Target tests use a verified sender, authorized real inbox and approved redirect URLs. Verify signup, confirmation, login, expired recovery, valid recovery, password change and old-password rejection. Check that unknown-address responses do not expose membership. Record actual delivery timestamps and failures. Current public-email tests are blocked by the owner's lack of a sender domain.

TC 011 Legal engine

Maps FR 024 to FR 026. Target tests cover category inclusion/exclusion, effective-date boundaries, amendment precedence, packaging exceptions and unknown applicability. Every conclusion resolves to an approved source hash and exact clause. Withdrawn or draft rules cannot execute. Changed packs flag affected old reports without changing their recorded history. A legal expert approves fixtures before production use.

TC 012 Privacy and operations

Maps FR 023, FR 027, FR 028 and NFR 005 to NFR 007. Target tests switch accounts without leaking cached cases, revoke membership, delete eligible records and all associated objects, expire shares, and confirm documented backup handling. Restore an isolated backup and compare hashes and counts. Trigger a real redacted processing failure and prove the configured alert reached its operator. These are not satisfied by a reminder or local log.

Real packaging evaluation protocol

Collect rights-cleared Indian labels with consent or an appropriate public-use basis. Maintain separate train/tuning and held-out sets split by product or brand family to avoid duplicate leakage. Include flat boxes, curved bottles, reflective film, small inkjet dates, embossed text, partial panels and varied lighting. Include English, Hindi and each additional supported language in separately reported strata.

Annotators mark declaration text, image boxes, legibility, relevant surface coverage, conflicts and applicability. Two independent annotators adjudicate disagreements; a legal reviewer validates potential absence cases. Record the source and image rights. Never fabricate a benchmark or label an unphotographed declaration as missing.

Report per-field extraction precision/recall, normalized value accuracy, box localization, unreadability recall, conflict recall and false absence rate with counts and intervals. Proposed rollout gates are zero false absence findings in the accepted test set, evidence linkage for every observed finding, and expert-approved minimum extraction thresholds per supported stratum. These thresholds need owner approval and sufficient sample sizes; zero test failures is not a population guarantee.

Performance and accessibility tests

Measure on 390-pixel mobile and 1440-pixel desktop views, with representative uploaded images and cold OCR assets. Record hardware, network profile, bundle size, LCP, INP, CLS, memory use and OCR duration. Targets appear in SRS; current build gzip size is a useful artifact, not a field performance result. Test low storage quota, offline shell, reconnect and service-worker release updates.

Use automated accessibility checks plus keyboard and screen-reader review. Verify labels, landmarks, focus order, modal return focus, contrast, live status, 200 percent zoom and errors tied to inputs. Broader Safari and Firefox validation remains pending until executed. Do not claim WCAG certification from semantic HTML alone.

Defects and release gates

Critical defects include cross-tenant exposure, privilege escalation, silently overwritten evidence and unsupported automatic absence conclusions. High defects include loss of originals, broken exports or a processing recovery path that duplicates cases. Fix these before the affected feature ships. Record lower-severity problems with owner and target release.

For the current prototype, require unit, full local workflow, document downloads, build, deployment and public-demo checks. For production, also require public-email evidence, all legal and privacy Target cases, a held-out benchmark, job recovery, restore and alert drills. Tag every unexecuted case Pending, Blocked or Not applicable with a reason. The final verification report lists the actual commands executed on 6 October 2026 and must be updated after the last deployment.